

ORIGINAL INVESTIGATION

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CT findings in five dogs with surgically confirmed colonic torsion

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Abstract

Colonic torsion is a life-threatening condition that results in colonic ischemia, necrosis, perforation, sepsis, and eventual death. The aim of this multicenter, retrospective case series study was to describe the CT findings in dogs with surgically confirmed colonic torsion. Medical records were searched for dogs with surgically confirmed colonic torsion following abdominal CT. Five dogs met the inclusion criteria. Three had a history of chronic intermittent diarrhea prior to presentation. Two dogs presented with acute vomiting, diarrhea, and abdominal pain and one dog presented with acute vomiting and lethargy. Computed tomographic findings in all dogs with surgically confirmed colonic torsion include: "whirl sign," displacement and distension of the cecum and colon, focal narrowing of the colon, and distension of the mesenteric vasculature in all dogs (5/5); streaky peritoneal fat and peritoneal effusion (4/5), pneumatosis coli (2/5), small intestinal distension (2/5), portal vein thrombosis (1/5), and reduced colonic wall contrast enhancement (1/5). In all dogs (5/5), the torsion site was the descending colon and demonstrated an anticlockwise rotation. At surgery, three of the five dogs had a partial colonic torsion with hyperemia at the site of obstruction and two of the five dogs had a complete torsion with marked necrosis of the colonic wall. Displacement of the colon and cecum, segmental distension and focal narrowing of the colon, the presence of a "whirl sign" and distension of the mesenteric vasculature are CT findings highly suggestive of colonic torsion.

KEYWORDS

canine, colonic torsion, computed tomography, whirl sign

1 | INTRODUCTION

Colonic torsion, as in small or large bowel volvulus, is a life-threatening condition that results in colonic ischemia, necrosis, perforation, sepsis, and eventual death.^{1,4} Colonic torsion is defined as a twisting of the colon around its longitudinal axis. Colonic volvulus refers to rotation of the organ around its mesenteric pedicle.¹ Both conditions can occur simultaneously and these terms have been used interchangeably in the veterinary literature.^{2,3} Clinical signs are nonspecific and include acute or per-acute onset of vomiting, lethargy, abdominal discomfort, and hypovolemic shock.⁵ Other clinical signs such as diarrhea, tenesmus, and anorexia have also been reported.⁵ It has been

suggested that dogs with prior history of gastrointestinal disease and/or abdominal surgery may be predisposed to developing colonic torsion.^{4,6}

The radiographic signs of colonic torsion in dogs have recently been described.² The most common radiographic signs include segmental distension and focal narrowing of the colon and displacement of the descending colon and cecum. A "torsion sign," described as helical striations within the colonic wall, may also be present following the administration of a barium enema.² A CT "whirl sign" has been considered a specific feature of organ torsion or volvulus as it reflects the spiral path of the twisted tissue and associated blood vessels.⁷ It has been previously reported in dogs with splenic torsion, ileoceocolic volvulus, mesenteric volvulus, urinary bladder torsion, and testicular torsion.^{7–11} Published studies describing CT characteristics of colonic torsion in dogs were lacking at the time of the current study.

The aim of the current study was to describe CT findings in dogs with surgically confirmed colonic torsion.

EQUATOR network guideline disclosure: This study conforms to the CARE reporting guidelines.

Previous publication or presentation disclosure: Presented by Dr. Pablo Barge as an abstract during the Spring EAVDI-BID meeting held on Wednesday April 3rd 2019 in Birmingham (UK).

2 | METHODS

The study was a multicenter, retrospective case series design. The medical records of Pride Veterinary Centre (Derby), VetCT Consultants in Telemedicine (Cambridge), Dick White Referrals (Newmarket), The Queen Mother Hospital for Animals (Royal Veterinary College, London), Langford Vets Small Animal Referral Hospital (Bristol University, Bristol), The University of Edinburgh Hospital for Small Animals (Edinburgh), University College Dublin Veterinary Hospital (Dublin) and the Small Animal Teaching Hospital of the University of Liverpool (Liverpool), were searched for dogs with surgically confirmed colonic torsion following an abdominal CT. Owner consent was provided for all diagnostics and treatment and the subsequent use of all generated data.

Recording and analysis of medical data was supervised by an ECVDI certified radiologist (CF). The following information was recorded by a diagnostic imaging intern (PB): signalment, presenting clinical signs, physical examination findings at presentation, laboratory findings at presentation, any history of prior gastrointestinal disease, and surgical findings.

The CT studies were independently reviewed by two ECVDI certified radiologists (IJ and CF) who were aware of the final diagnosis at the time of interpretation. The CT studies were evaluated for the following criteria: segmental distension of the colon, focal narrowing of the colon, displacement of the cecum and colon, enhancement of the colonic wall, the presence or absence of pneumatosis coli, the presence or absence of a "whirl sign," small intestinal distension, the presence or absence of streaky peritoneal fat, peritoneal effusion, distension of the colonic and mesenteric vasculature and thrombosis. If present, the site of thrombosis was also recorded. The direction (clockwise/anticlockwise) and site of rotation were also recorded. Any discrepancies were resolved by consensus.

Segmental distension of the colon was defined as visualization of two different populations of colon. Distension of the colon was defined as a serosa to serosa thickness greater than the total length of L7.¹² Focal narrowing of the colon was defined as the presence of a sudden reduction in diameter of the colon.² Displacement of the cecum and ascending colon, the transverse colon and the descending colon, was defined as the presence of these structures in an abnormal location.¹³ Enhancement of the colonic wall following contrast administration was assessed subjectively as compared with that of the ileum.¹⁴ The presence of pneumatosis coli was defined as visualization of gas within the wall of the colon.¹⁵ The presence of a "whirl" sign was defined as visualization of a spiral pattern of soft-tissue and fat attenuation within the peritoneal cavity associated with the focal narrowing of the colon, which was assessed in transverse plane.¹⁶ Small intestinal distension was defined as a serosa to serosa thickness greater than 1.6 times the height of L5, adapted from reported methods of radiographic assessment.¹⁷ The direction of rotation was based on assessing the images from caudal to cranial in a transverse plane. Additional findings were also recorded if considered relevant (eg, lymphadenopathy). The CT findings were then compared to the surgical findings.

3 | RESULTS

Five dogs met the inclusion criteria, two Great Danes, two Labrador Retrievers, and one German Shepherd. All were male: three were neutered and two were intact. Ages ranged from 5.0 to 7.8 years (mean 6.7 years). Weight ranged from 26 to 112 kg (mean 43 kg). Three dogs had a history of chronic intermittent diarrhea prior to presentation. None of the dogs had previously undergone abdominal surgery and/or gastropexy. Two dogs presented with acute (<24 h) vomiting, diarrhea, and abdominal pain. One dog presented with acute vomiting and lethargy. Two dogs presented with subacute clinical signs (up to 5 days duration) including anorexia, diarrhea, and vomiting. Physical examination revealed tachycardia in two dogs, abdominal pain in one dog, pyrexia in one dog, and was considered unremarkable in one dog. Signalment, previous medical history, presenting clinical signs, physical examination findings at presentation, surgical findings, and outcome are summarized in Supplement 1, Supporting Information.

Hematological and biochemical analysis at presentation was available in four dogs. It revealed a mild to moderate neutrophilia in two dogs, mild polycythemia in one dog, and mild anemia in one dog. Biochemistry revealed mild increases in aspartate aminotransferase and creatine kinase in three dogs and a mild increase in alanine aminotransferase in one dog.

Computed tomographic studies were performed using 16-slice CT scanners (General Electric Medical Systems, BrightSpeed or Lightspeed, Chicago, IL, USA), helical scan mode, 1.25–2.5 mm slice thickness, 1.25–2.5 mm reconstruction interval, and a pitch of 0.9–1.75 using the following technique: 104–336 mAs tube current, 120 kV tube voltage, 265–500 mm field of view, 512 × 512 matrix, and a soft tissue reconstruction algorithm. Window width and level were adjusted accordingly and three-dimensional and multi-planar reformatting were used whenever necessary for optimal evaluation. For all dogs, CT images were acquired prior to and at approximately 45 s following administration of iohexol (Omnipaque, Oslo, Norway, 300 mgI/mL) at 600 mgI/kg and at an injection rate of 4–5 mL/s via a cephalic catheter by an angiographic pressure injector (Nemoto Dual Shot Alpha 7, Tokyo, Japan).

The CT results for individual dogs are summarized in Supplement 2, Supporting Information, which compares dogs with partial versus complete torsion. All dogs (5/5) showed segmental distension and focal narrowing of the colon and displacement of the cecum and colon (Figure 1). Colonic wall enhancement was considered to be reduced in one dog (Dog 5) (Figure 2). A "whirl sign" was present in all dogs (5/5, Figure 3). Two dogs (2/5) had evidence of pneumatosis coli and small intestinal distension. Peritoneal effusion and streaky peritoneal fat were present in four dogs (4/5). The colonic and mesenteric vasculature were also considered distended in all dogs (5/5; Figure 4), whereas portal vein thrombosis was only identified in Dog 5 (1/5) at the level of the *porta hepatis*. In addition to the "whirl sign," all dogs (5/5) had evidence of an axial, anticlockwise rotation of the collapsed descending colon aborad to the site of obstruction (Figure 5), which was located in the descending colon in all cases (5/5). Additional

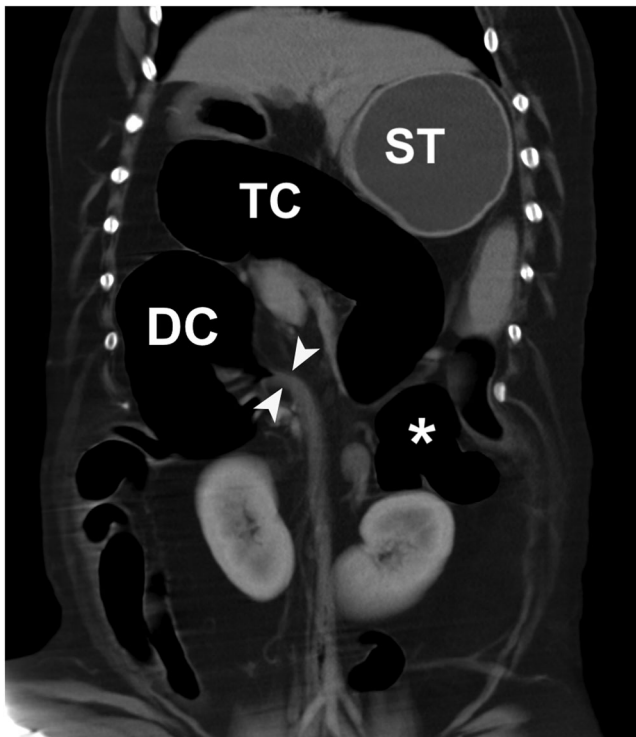


FIGURE 1 Postcontrast, soft tissue (window level = 90 and window width = 650) dorsal maximum intensity projection reconstruction (12 mm) of the abdomen of Dog 1. The arrowheads indicate focal narrowing of the colon. The cecum (asterisk) is displaced and is located on the left. The descending colon is also displaced and is located on the right
Abbreviations: DC, Descending colon; ST, stomach; TC, transverse colon



FIGURE 2 Postcontrast, soft tissue (window level = 90 and window width = 650) transverse image of the mid-abdomen at the level of the caudal pole of the left kidney (+) of Dog 5. Note the poor enhancement of the wall of the displaced descending colon (arrowheads) and of the displaced cecum (asterisk) as compared with the small intestine (SI). There is a small amount of mineralized material within the dependent aspect of the descending colon and peritoneal effusion



FIGURE 3 Postcontrast, soft tissue (window level = 90 and window width = 650) transverse maximum intensity projection reconstruction (12 mm) of the mid abdomen of Dog 1. A mixed, soft-tissue, and fat attenuating "whirl" is present, composed of the colonic vessels (arrowheads), surrounding the collapsed descending colon (arrow)

findings included a mild jejunal lymphadenopathy, which was present in all dogs (5/5).

Exploratory laparotomy revealed a partial colonic torsion in three dogs (3/5; Dogs 1, 3, and 4). There was evidence of colonic wall congestion and hyperemia at the site of obstruction that resolved following repositioning in all three dogs. Colopexy and prophylactic gastropexy were performed in these dogs. Two dogs were discharged after surgery and recovered uneventfully. One dog was euthanized 2 weeks after surgery due to lack of clinical improvement following discharge.

Exploratory laparotomy revealed a complete colonic torsion in two dogs (2/5; Dogs 2 and 5). Marked necrosis of the colonic wall was present in both these dogs. One dog was euthanized intraoperatively. An ileorectal anastomosis was performed in the other dog. This patient did not survive to discharge and was euthanized 6 days after surgery due to a lack of clinical improvement.

4 | DISCUSSION

The use of CT was preferred in our dogs due to their large size and the ease of image acquisition compared to other modalities such as radiography or ultrasound.^{18,19} It also offers additional advantages over radiography such as superior contrast resolution without superimposition. In addition, CT may help diagnose potential sequel to colonic torsion that cannot be identified using radiography alone. The possible presence of peritoneal effusion may also reduce abdominal serosal detail, further limiting radiographic assessment. The authors recommend the use of CT to diagnose canine colonic torsion in large breed dogs.

The current study findings are in agreement with previous studies stating that colonic torsion is a rare condition, typically seen in

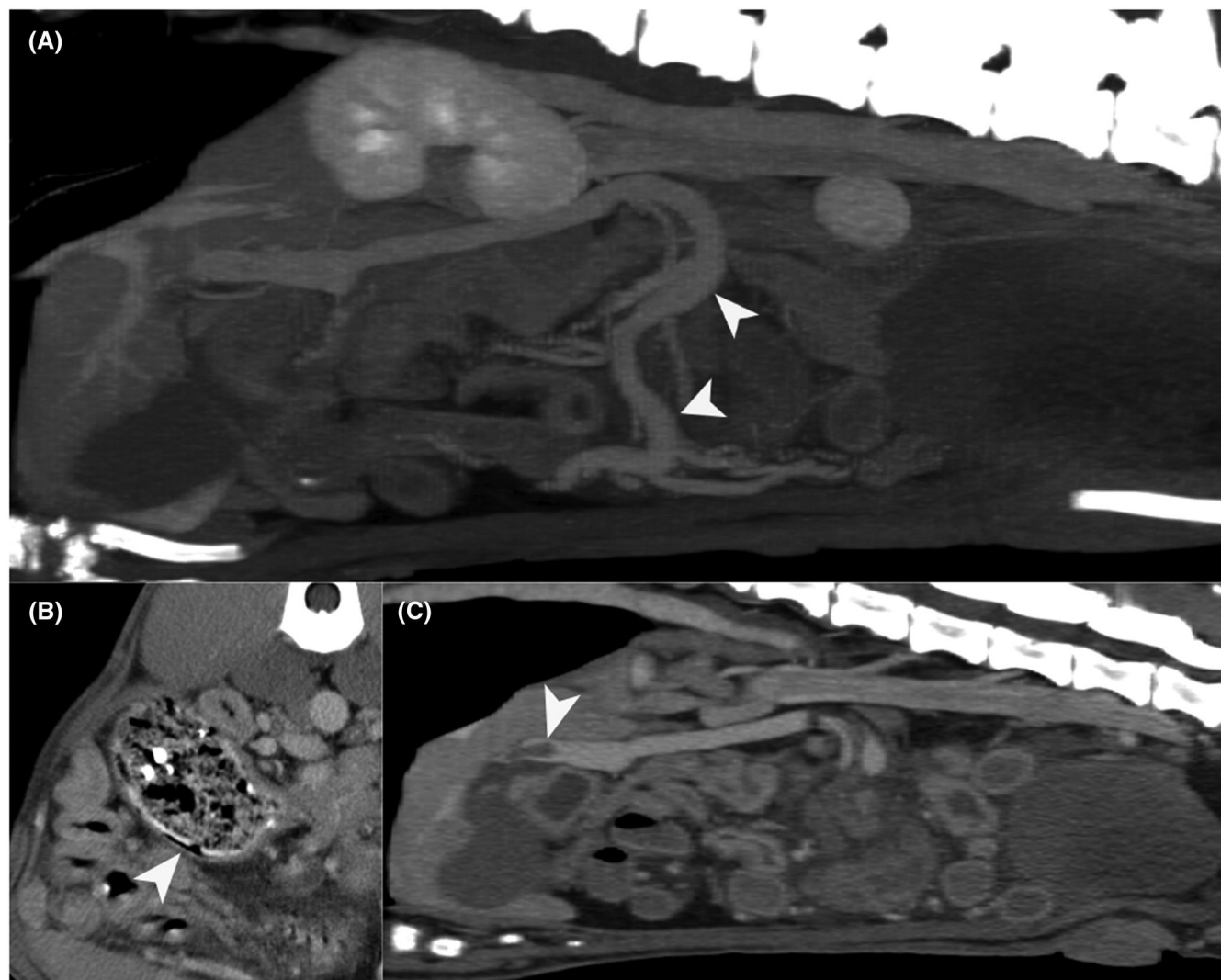


FIGURE 4 A, Postcontrast, soft tissue (window level = 90 and window width = 650) sagittal maximum intensity projection reconstruction (20 mm) of the abdomen of Dog 5. There is severe distension of the mesenteric veins (arrowheads). B, Transverse view of the mid-abdomen in a soft tissue window (window level = 90 and window width = 650) at the level of L2 of Dog 4. Gas is identified tracking along the ventral aspect of the wall of the descending colon aborad to the site of torsion (arrowhead). C, Sagittal view of the mid abdomen in soft tissue window (window level = 90 and window width = 650) of Dog 5. A small filling defect is identified in the portal vein at the *porta hepatis* (arrowhead)

adult, male, large breed dogs such as Great Danes, Labrador Retrievers, and German Shepherds.^{2,4,6,20} The reason for a higher incidence in males remains unknown.²¹ A predisposition to colonic torsion has been suggested in dogs with a history of previous abdominal surgery and/or chronic gastrointestinal disease. In one previous study with six dogs with colonic torsion/volvulus, four dogs had a prior history of gastric dilation-volvulus with prophylactic gastropexy.^{5,6} None of the dogs in the current study had a history of previous abdominal surgery. Three of the five dogs in this study had a history of chronic intermittent diarrhea. Similar clinical history has previously been described in other studies.²⁰ Predisposing factors for colonic torsion remain largely unknown.

Large dogs may be predisposed to colonic torsion due to specific anatomic differences such as the large size of their peritoneal cavity relative to the volume of their colon. As reported previously, large

body size and deep-chested dogs of certain breeds are predisposed to gastric dilation-volvulus.²² A markedly distended stomach in a large abdominal cavity may stretch the hepatogastric ligament, increasing gastric mobility. This ligament may also progressively stretch and become less compliant as the dog gets older.²² Similar changes to the mesocolon may increase colonic mobility resulting in a predisposition to colonic torsion. Certain breeds such as German Shepherds are also known to demonstrate higher incidence of gastrointestinal displacement disorders.²⁰

Clinical signs in dogs with colonic torsion are usually acute and nonspecific and most commonly include vomiting, lethargy, abdominal pain, and hypovolemic shock.^{2,4,5,20} Dogs may also present with diarrhea or tenesmus, although this is considered to be less common.⁵ This is supported by the results of the current study. None of the dogs with confirmed colonic torsion in this study were considered to be in

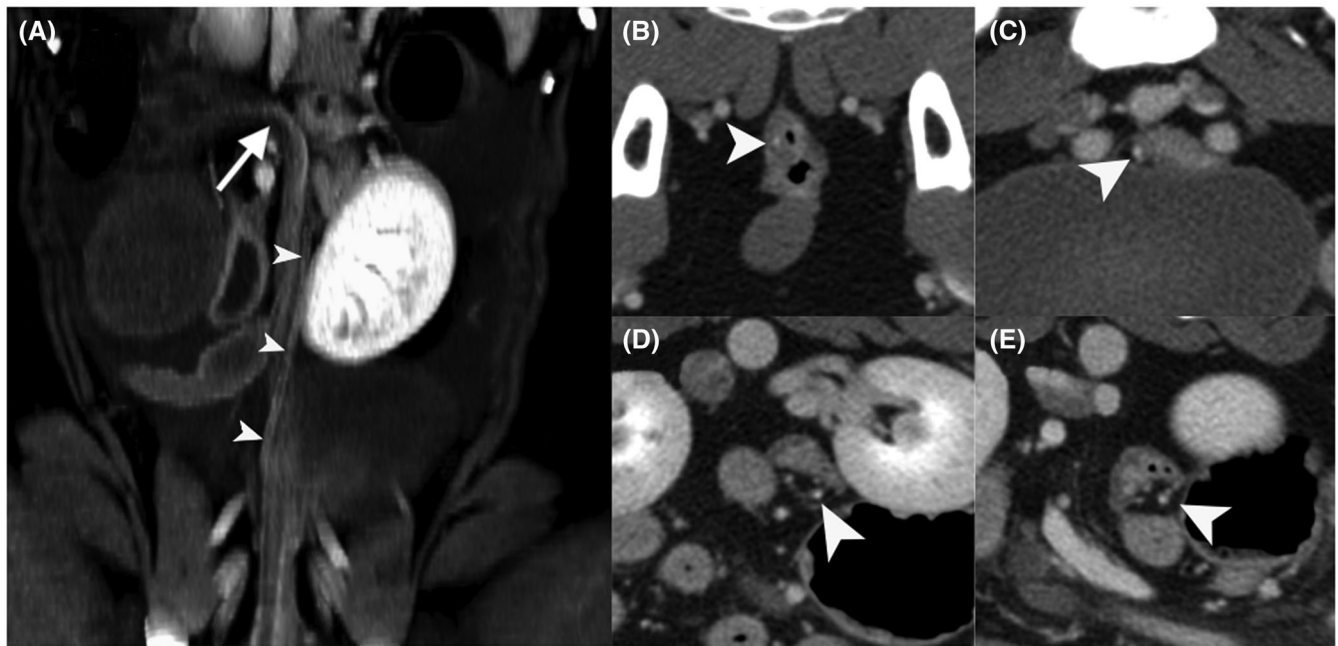


FIGURE 5 A, Postcontrast soft tissue (window level = 90 and window width = 650) dorsal maximum intensity projection reconstruction (12 mm) of the caudal abdomen of Dog 5. The left colic vein/cranial rectal veins are encircling the collapsed descending colon along its entire length (arrowheads). The arrow indicates the location of the torsion. B-E, Transverse postcontrast soft tissue reconstructions (window level = 90 and window width = 650) of Dog 3 from caudal to cranial. Note the anticlockwise rotation of the collapsed descending colon highlighted by the encircling left colic/cranial rectal vein (arrowheads)

hypovolemic shock at the time of presentation. This may be due to the fact that none of the dogs had impeded venous return to the heart as is often the case in dogs with gastric dilation-volvulus.²³ In dogs with colonic torsion, the shock is more likely to be triggered by the ischemic injury of the colonic wall²⁴ and therefore occur much later relative to the onset of clinical signs.

The most common CT findings in our study included segmental distension of the colon, focal narrowing of the colon, displacement of the cecum and colon, a “whirl sign,” and vascular distension. These findings were identified in all dogs in this study and were most likely caused by rotation of the colon around its longitudinal axis and associated colonic and vascular obstruction. The “whirl sign” is specific to CT evaluation and cannot be identified using either radiographs or ultrasound. It has been previously described in different conditions^{7–11} and it is also considered a key feature of bowel volvulus in human patients.¹⁶ The authors suggest that the “whirl sign” may be used to aid in the diagnosis of canine colonic torsion. However, a recent case report described the CT findings in a dog with colonic torsion and entrapment due to an omental rent and included segmental distension, displacement and focal narrowing of the colon, and an approximate 180° anticlockwise axial rotation of the descending colon.¹⁸ A “whirl sign” and mesenteric vascular distension were not identified on CT and the torsion was not recognized during surgery.¹⁸ Dynamic colonic torsion has also been suggested by other authors² and it is possible that a “whirl sign” may not be present in these cases. Therefore the absence of a “whirl sign” does not completely rule out the presence of a colonic torsion.

The most common site of colonic torsion in our study was the descending colon (5/5). This may be due to increased compliance of the

ligament/mesocolon in this region. An anticlockwise rotation occurred in all dogs (5/5) when viewed caudal to cranial. This is similar to dogs with a gastric dilation-volvulus where the pylorus shifts in an anticlockwise direction from a ventral position on the right, to a dorsal position on the left.²⁵ Given the location, shape and position of the large bowel, and the anticlockwise rotation identified in all of our cases, it is possible that the transverse colon effectively acts as a sling, with the cecum acting as the load, which then rotates around the long axis of the descending colon. The cecum first moving ventrally into the gravity-dependent portion of the abdomen and then propelled dorsally and to the left by generated centripetal force, rotating around the long axis of the descending colon in an anticlockwise direction. It is likely that a combination of conditions is required in order for this to happen, considering the rarity of this condition.

Only two dogs demonstrated small intestinal distension, whereas all dogs demonstrated segmental distension of the colon. This is likely due to the location of obstruction associated with the torsion. The bowel immediately orad to an obstruction typically becomes dilated. As colonic torsion may result in a colonic obstruction, it is the large bowel that is most likely to become dilated.

Other CT findings in our study included streaky peritoneal fat and peritoneal effusion, identified in four dogs. The increased intraluminal pressure of the affected segment leads to a compromise of the mucosal microcirculation, which contributes to bowel ischemia and ultimately bacterial overgrowth and necrosis.²⁶ Bacterial and fluid translocation can lead to peritoneal and endothelial damage²⁴ eventually resulting in peritoneal effusion and inflammation visible on CT as streaky hyperattenuation of the peritoneal fat. The authors consider that these

findings may be directly related to the severity and duration of the condition. Thrombosis was seen in only one dog, presumably as a combination of reduced blood flow and associated endothelial injury.²⁴

Pneumatosis coli was present in two dogs, neither of which demonstrated signs of colonic wall necrosis at surgery. Gastric pneumatosis may occur secondary to gastric dilation-volvulus due to mucosal tearing as a result of increased intraluminal pressure, rather than as a result of necrosis.^{27,28} The same pathophysiological mechanisms may occur in cases of colonic torsion.

Only one dog was considered to have poor enhancement of the colonic wall (Dog 5). At surgery, this dog had extensive necrosis of the colonic wall and a complete torsion. However, colonic wall necrosis was reported in another dog at surgery where colonic wall enhancement was considered normal on CT. In human patients, precontrast hyperattenuation of the colonic wall secondary to transmural hemorrhagic necrosis, colonic wall thickening, poor enhancement of the colonic wall, and pneumatosis coli are all findings that may be used to predict colonic wall viability.²⁹ Further investigations are required in order to investigate whether CT findings may help predict the degree of torsion and colonic wall viability in dogs with colonic torsion.

Three dogs with a partial torsion identified at surgery survived to discharge; however, one was euthanized 2 weeks after surgery. The two dogs with a complete colonic torsion identified at surgery did not survive to discharge (Supplement 1, Supporting Information). This is not surprising as the risk of vascular occlusion and associated ischemia of the colonic wall, subsequent necrosis, bacterial translocation, and eventual septic shock is likely to be increased in dogs with a complete versus a partial colonic torsion. Computed tomographic findings were similar in dogs with partial versus complete colonic torsion (Supplement 2, Supporting Information) and estimating the degree of torsion based solely on CT may not be possible.

The main limitation of our study is the small sample size. Only five dogs satisfied our inclusion criteria, despite the inclusion of eight veterinary hospitals. This supports previous evidence suggesting that colonic torsion is a rare condition in dogs.² It is also possible that in many cases of colonic torsion, CT is not required as radiographs can be used to achieve a diagnosis.² Other limitations of the study include the lack of histopathologic analysis to confirm the presence of colonic wall necrosis and/or pneumatosis coli.

In conclusion, authors recommend that the presence of segmental distension of the large intestine, focal narrowing of the colon, displacement of the colon and cecum, a "whirl sign," and distension of the mesenteric vasculature be considered as CT findings highly suggestive of colonic torsion.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Barge, Fina, Jones
- (b) Acquisition of Data: Barge, Fina, Mortier, Jones
- (c) Analysis and Interpretation of Data: Barge, Fina, Jones

Category 2

- (a) Drafting the Article: Barge, Fina, Jones
- (b) Revising Article for Intellectual Content: Barge, Fina, Mortier, Jones

Category 3

- (a) Final Approval of the Completed Article: Barge, Fina, Mortier, Jones

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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